

◆ 16.6 C2 Integration & Conflict Prevention (Final)

Core Principle

A single authoritative decision path governs all execution.

16.6.1 Unique Decision Constraint

$$d(t) \in \mathcal{D}, \quad \quad |\mathcal{D}_{\{\mathrm{active}\}}| = 1$$

16.6.2 Validation Gate

$$l_{\{\mathrm{eff}\}} \geq \theta_{\{\mathrm{ind}\}} \Rightarrow d(t) \text{ valid}$$

Else:

$$d(t) = \text{varnothing}$$

● 16.6.3 Conflict Resolution (Corrected)

بدل "reject both" (دي مش دقيقة رياضياً)، نستخدم:

$$d_1 \neq d_2$$

$\rightarrow$

$\gamma = 0$, re-evaluation

16.6.4 Execution Mapping

$$u(t) = \mathcal{C}_2(d(t))$$

16.6.5 Single Execution Path

$$u_{\text{final}}(t) = \begin{cases} u(t) & \text{if from } C_2 \\ 0 & \text{otherwise} \end{cases}$$

16.6.6 Runtime Safety Monitoring

$$\frac{d}{dt} I_{\text{eff}} < 0$$

$\rightarrow$
 $\gamma \rightarrow 0$

 Insight

Decision is enforced as a controlled pipeline, not a selection process.

16.7 System Metrics (Final — Formalized)

Core Principle

Only measurable quantities define system behavior.

Primary Metrics

$I_{\mathrm{eff}}(t), \quad \gamma(t), \quad Q(t),$
 $N(t)$

Operational Metrics

$$R_{\mathrm{act}} = \frac{\#\gamma=1}{\text{total attempts}}$$

$$R_{\mathrm{block}} = 1 - R_{\mathrm{act}}$$

$$T_{\mathrm{resp}} = t_{\mathrm{decision}} - t_{\mathrm{input}}$$

Stability Metric

$$\dot{V}(x) \leq 0$$

Consistency Metric (Important Fix)

لازم clamp علشان ما تبقاش سالبة:

$$C_{\mathrm{consistency}} = \max\left(0, 1 - (C_{\mathrm{struct}} + C_{\mathrm{time}} + C_{\mathrm{loop}})\right)$$

Error Rate

$E =$

$$\frac{\text{incorrect decisions}}{\text{total decisions}}$$

Final Insight

The system is fully observable, measurable, and verifiable.

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
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